

Classroom 50

Teacher Training Session

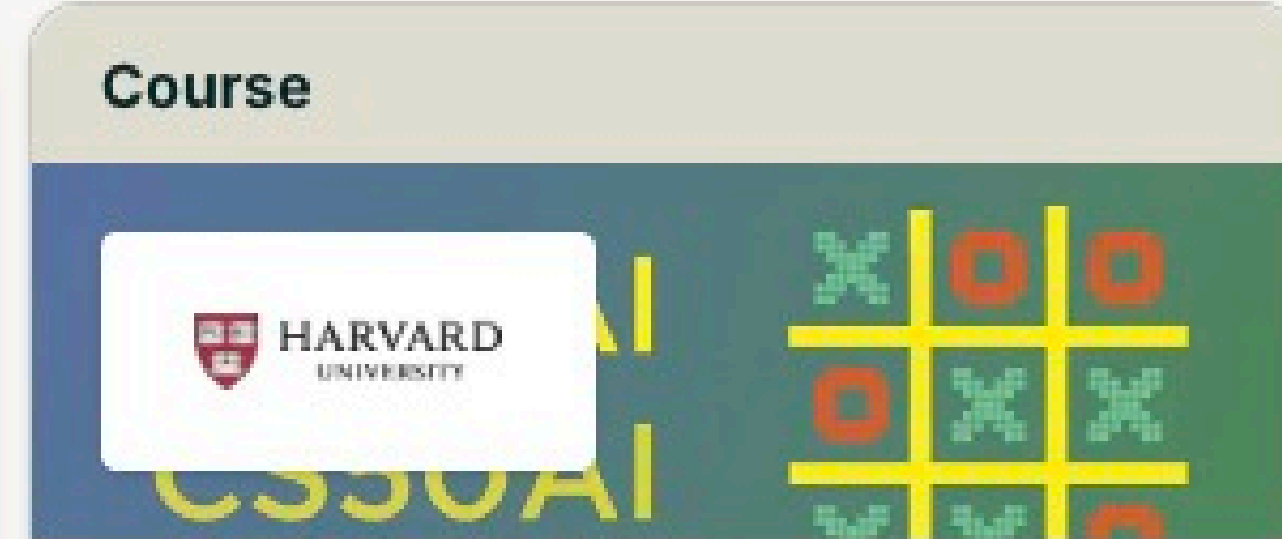
CS50

<https://edx.org/cs50>



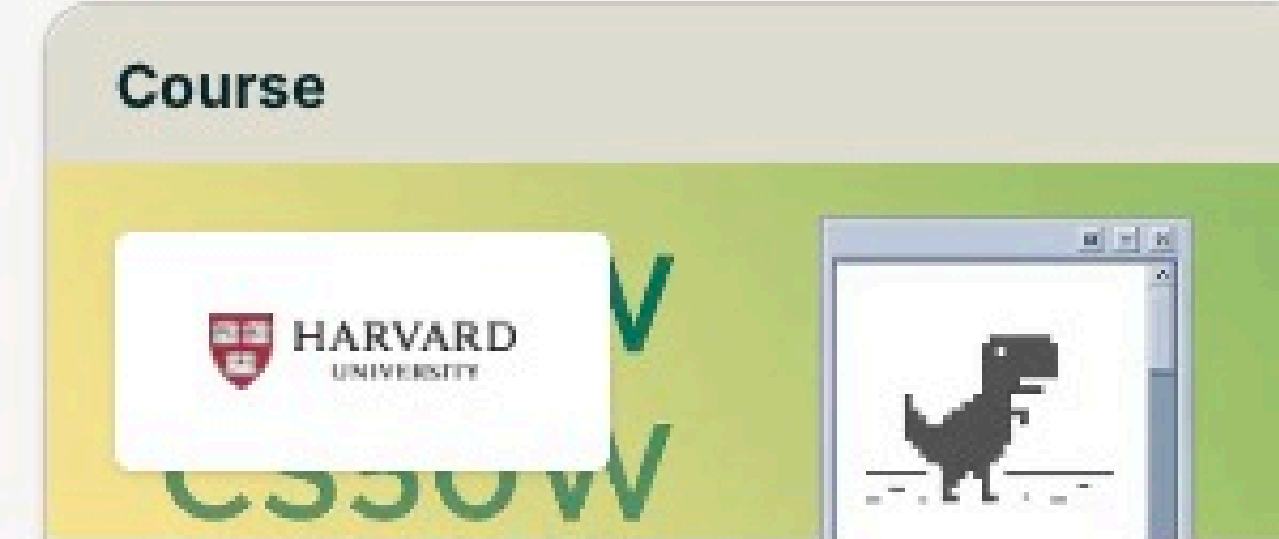
CS50's Introduction to Computer Science
Harvard University

🕒 3 months to complete
📖 Introductory level



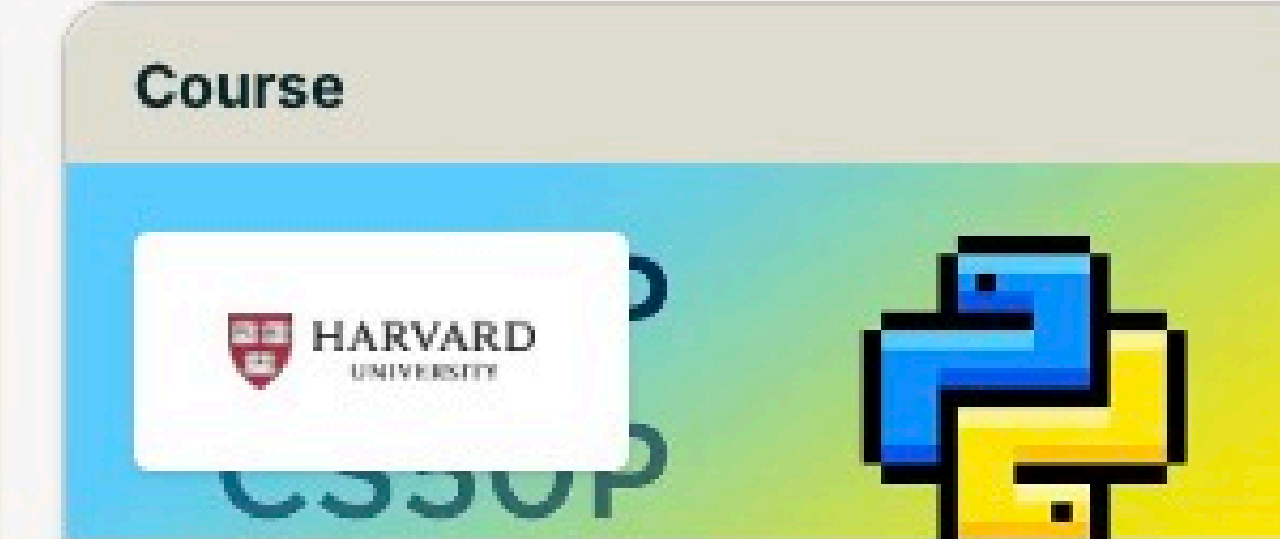
CS50's Introduction to Artificial Intelligence with Python
Harvard University

🕒 7 weeks to complete
📖 Introductory level



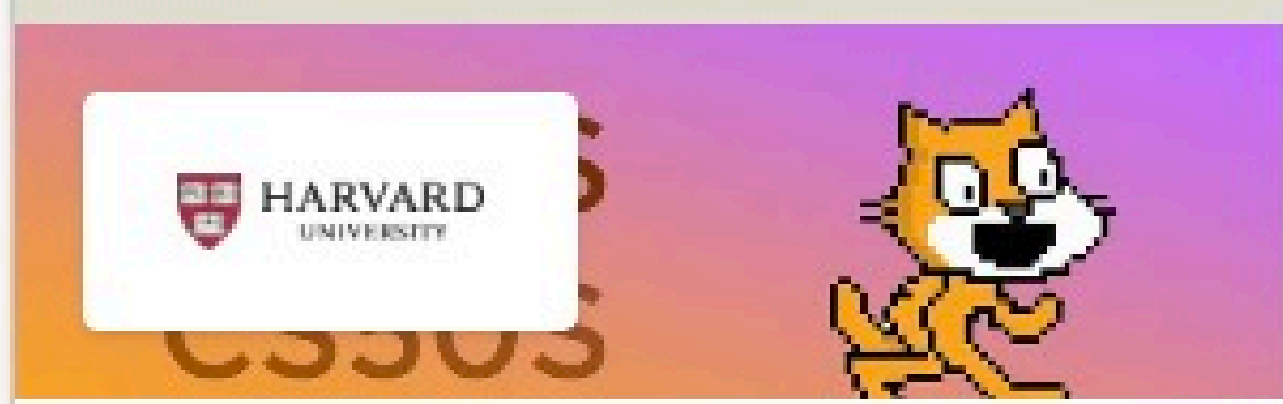
CS50's Web Programming with Python and JavaScript
Harvard University

🕒 3 months to complete
📖 Intermediate level



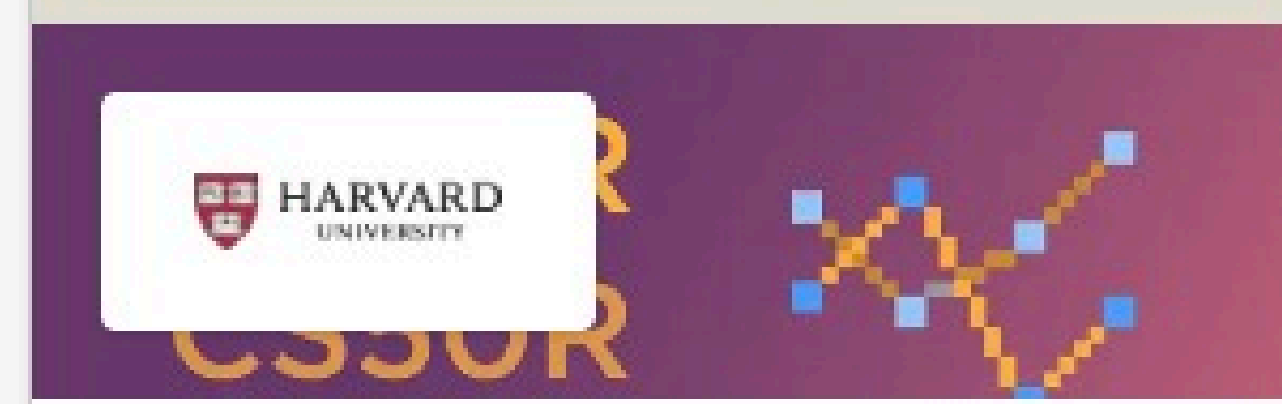
CS50's Introduction to Programming with Python
Harvard University

🕒 3 months to complete
📖 Introductory level



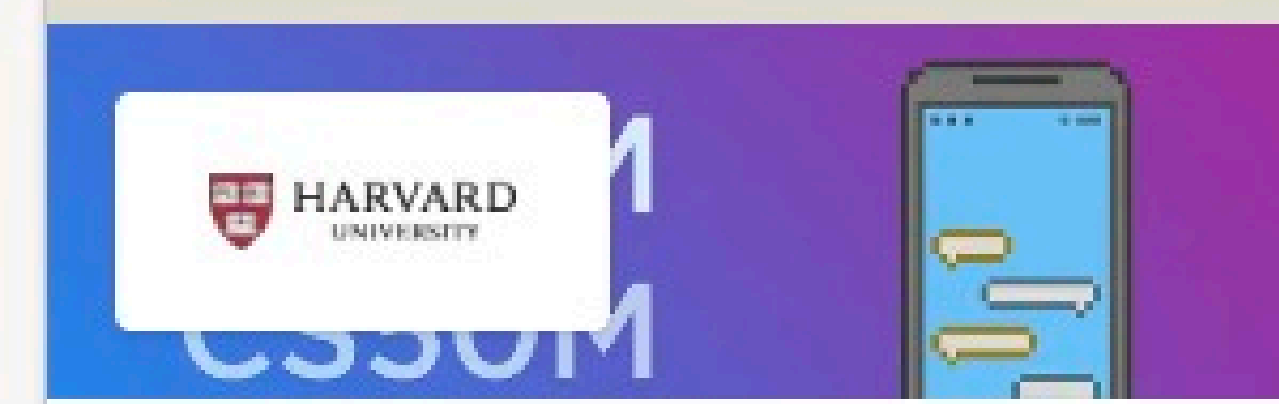
CS50's Introduction to Programming with Scratch
Harvard University

🕒 3 weeks to complete
📖 Introductory level



CS50's Introduction to Programming with R
Harvard University

🕒 7 weeks to complete
📖 Introductory level



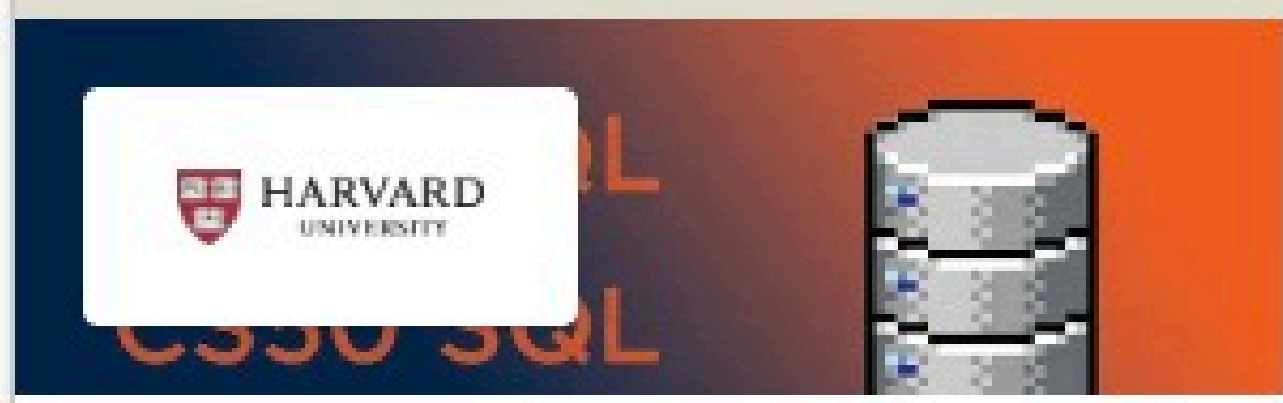
CS50's Mobile App Development with React Native
Harvard University

🕒 4 months to complete
📖 Intermediate level



CS50's Introduction to Cybersecurity
Harvard University

🕒 5 weeks to complete
📖 Introductory level



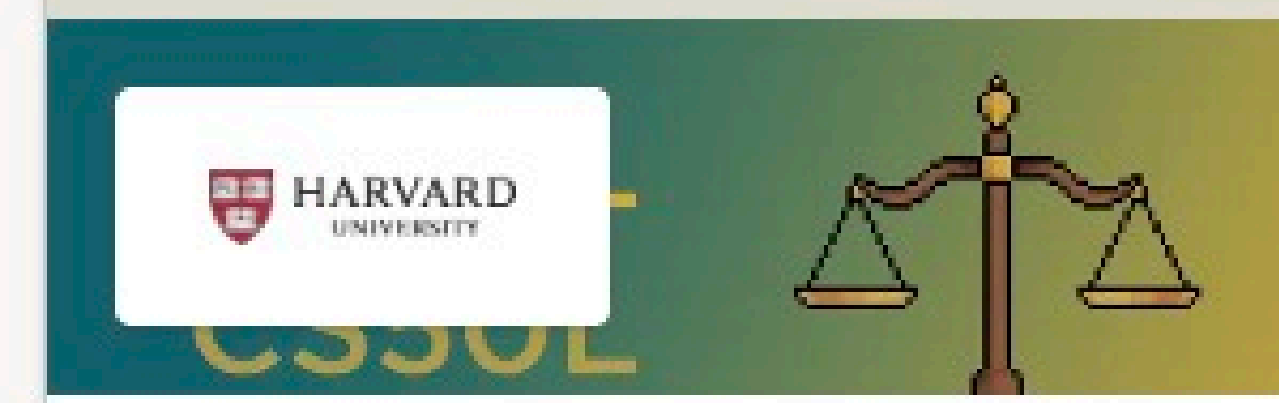
CS50's Introduction to Databases with SQL
Harvard University

🕒 7 weeks to complete
📖 Introductory level



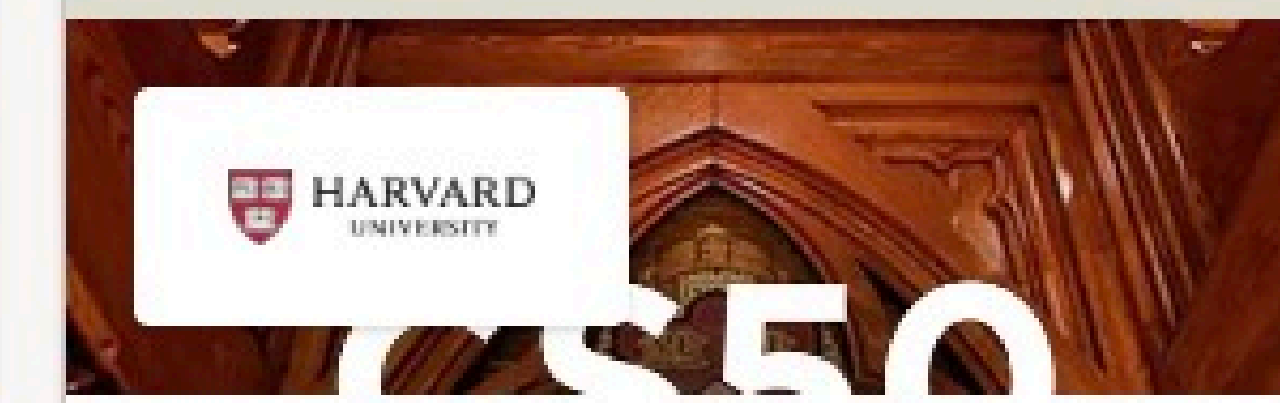
CS50's Computer Science for Business
Harvard University

🕒 8 months to complete
📖 Introductory level



CS50's Computer Science for Lawyers
Harvard University

🕒 3 months to complete
📖 Introductory level



CS50's AP Computer Science Principles
Harvard University

🕒 9 months to complete
📖 Introductory level

CS50 atop GitHub

- GitHub Classroom for **distributing assignments**
- GitHub Repositories for **storing submissions**
- GitHub Organizations for **organizing submissions**
- GitHub Teams for **managing permissions**
- GitHub Actions for **autograding**
- GitHub Codespaces for **writing code**
- GitHub OAuth for **authenticating**
- GitHub Container Registry for **storing Docker images**
- GitHub Copilot for **developing final projects**
- GitHub REST API for **automating tasks**

CS50 atop GitHub

- `submit50`
- `check50`
- ...

Fifty Foundation

<https://fifty.foundation>

Classroom 50

<https://classroom50.org>

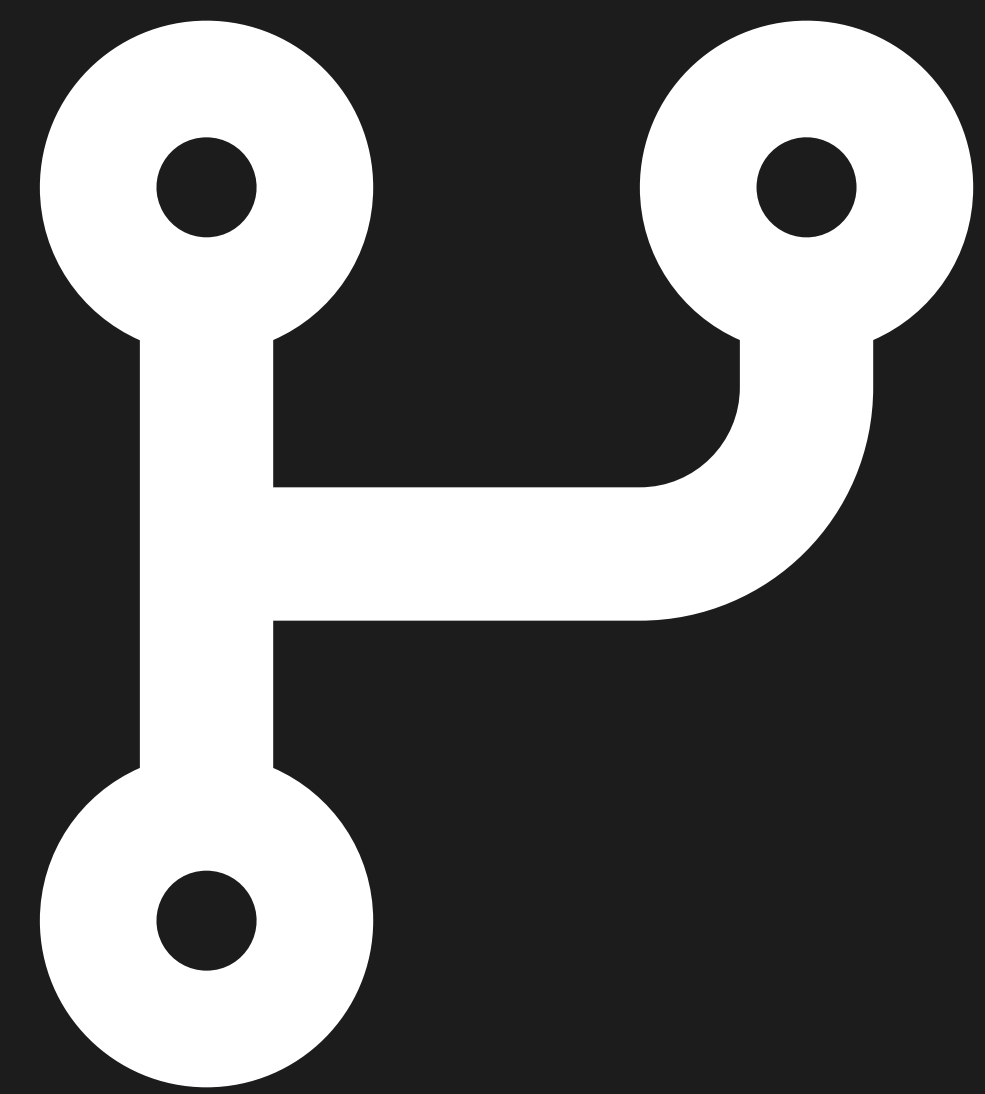
Classroom 50

<https://github.com/foundation50/classroom50>

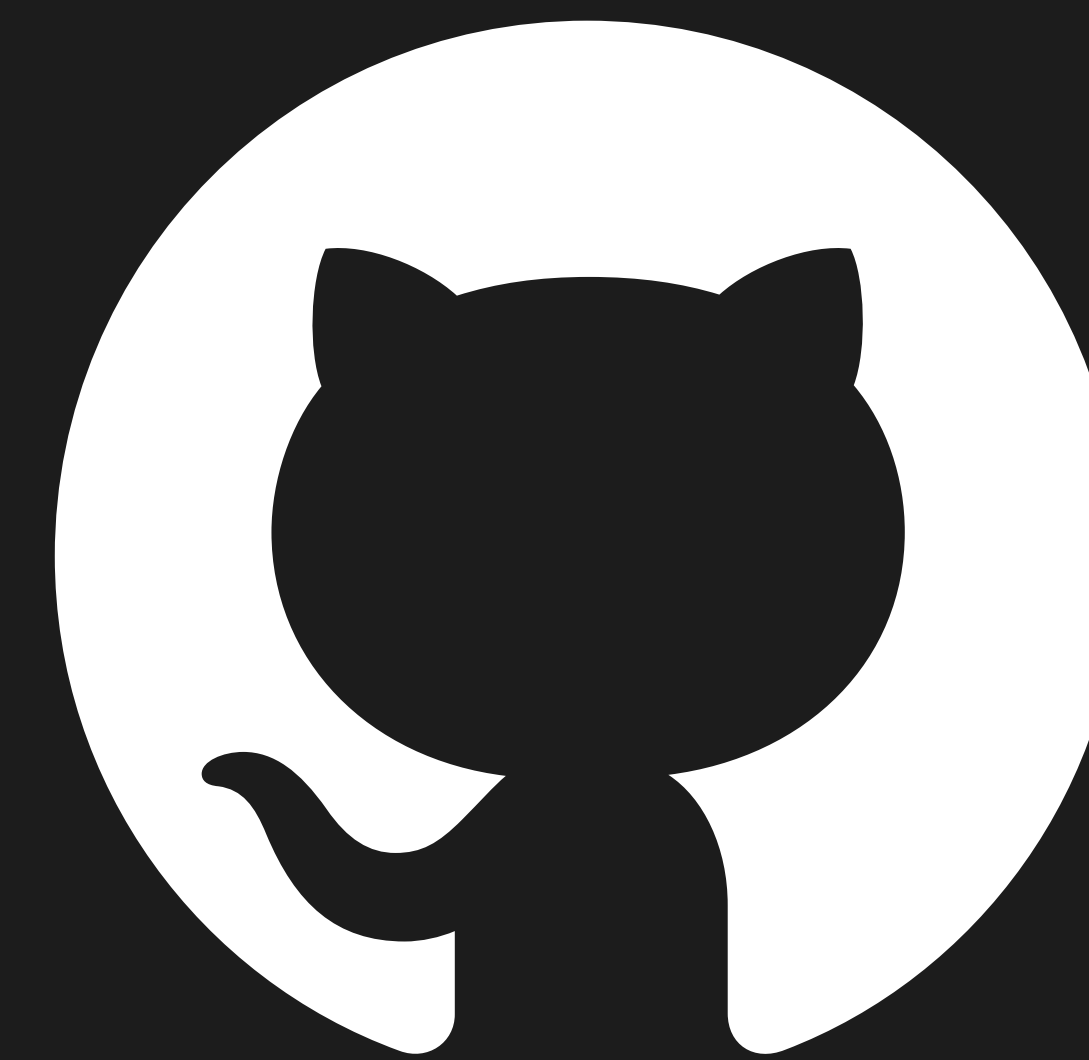
Classroom 50

- Create assignments with starter code
- Define correctness tests
- Distribute individual and group assignments
- Collect submissions
- Autograde submissions
- Provide inline qualitative feedback on submissions
- Manage course rosters and scores

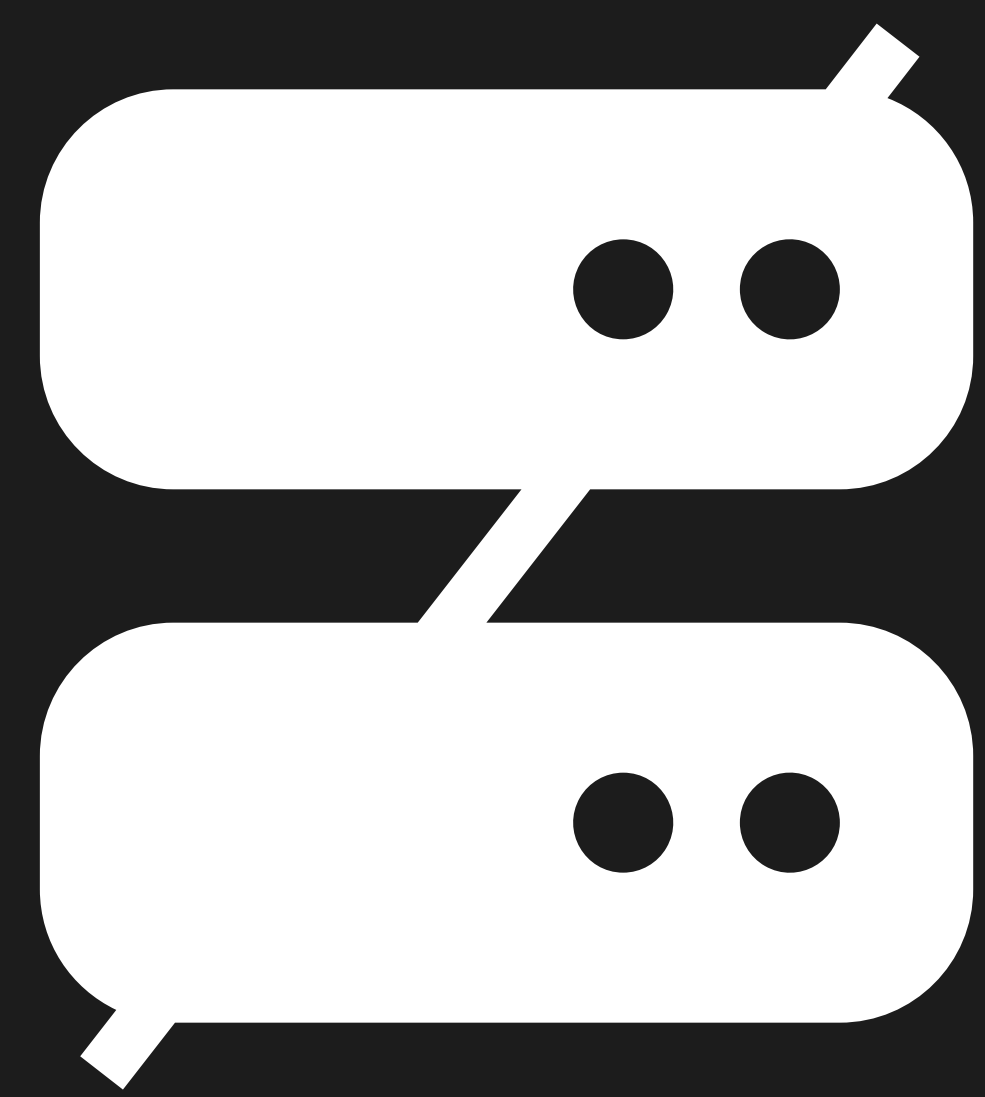
Design



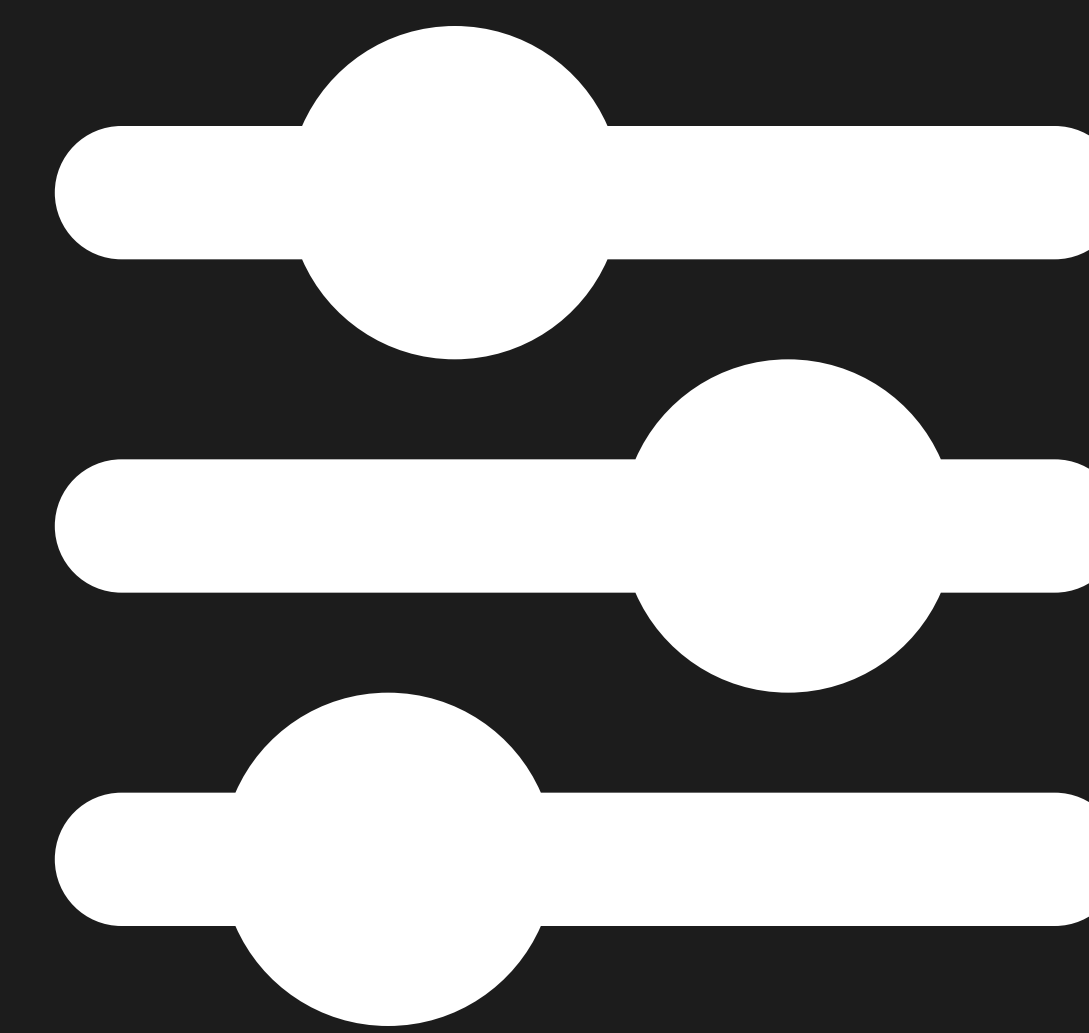
Open-Source



GitHub-Based

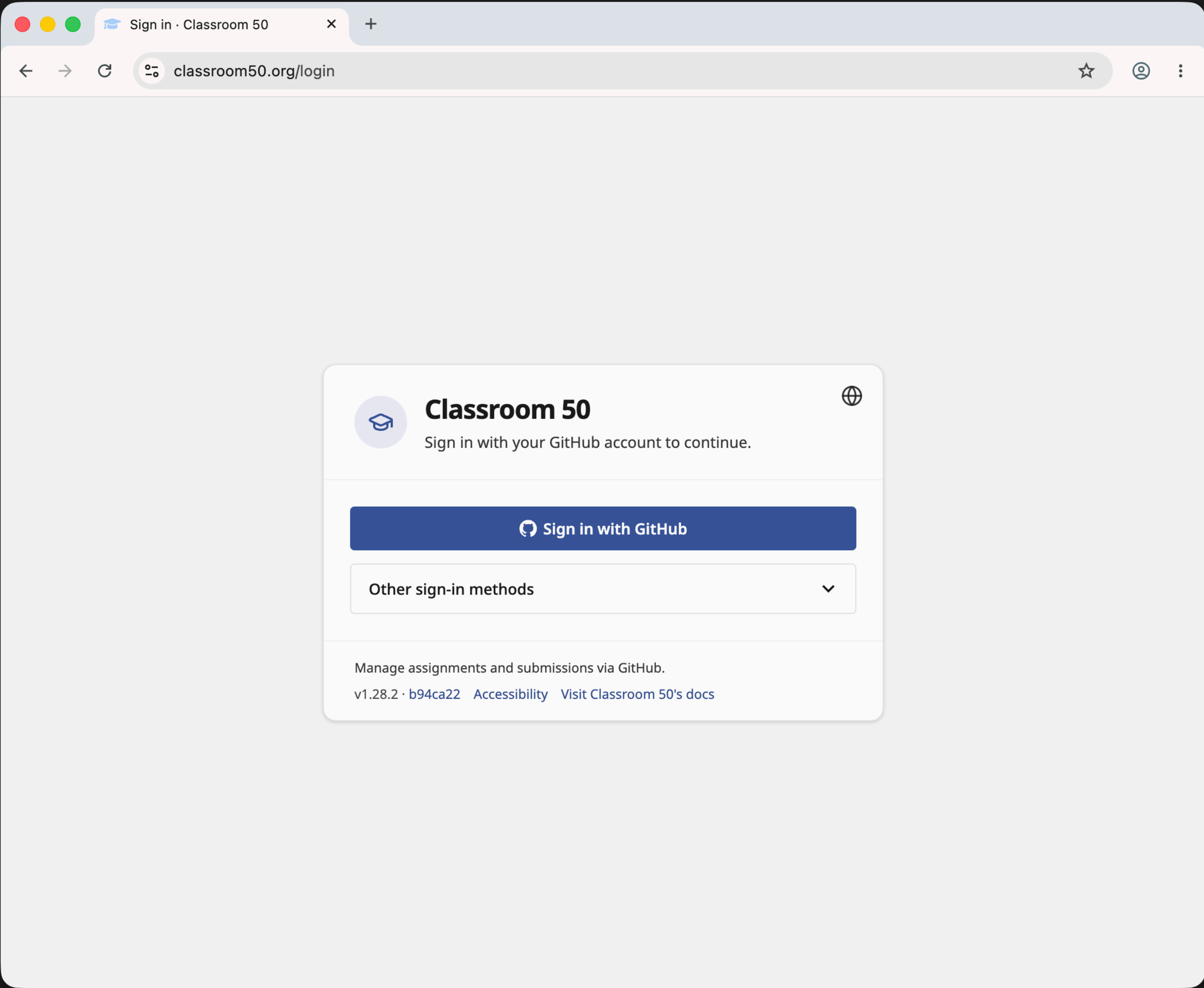


Serverless

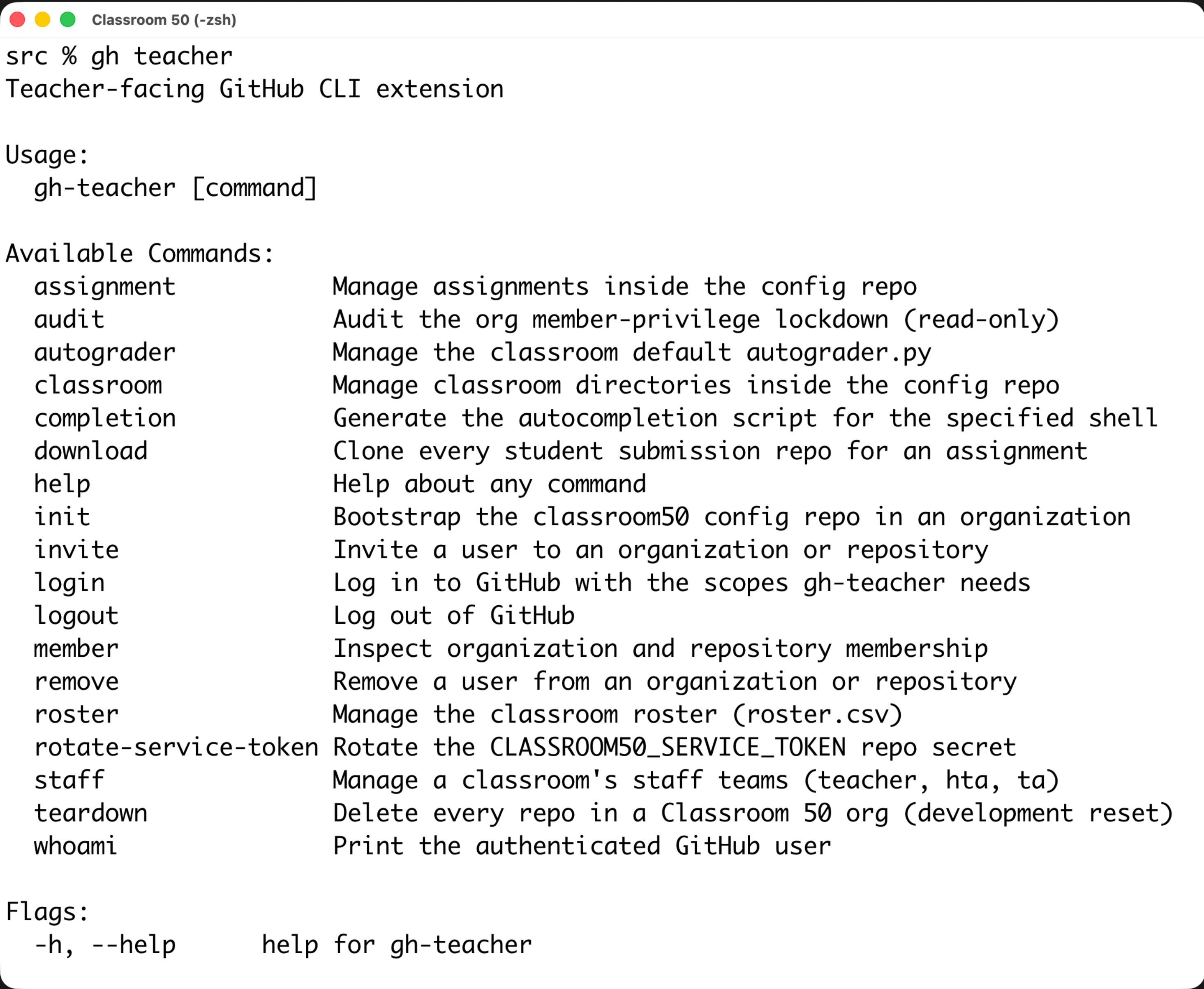


Customizable

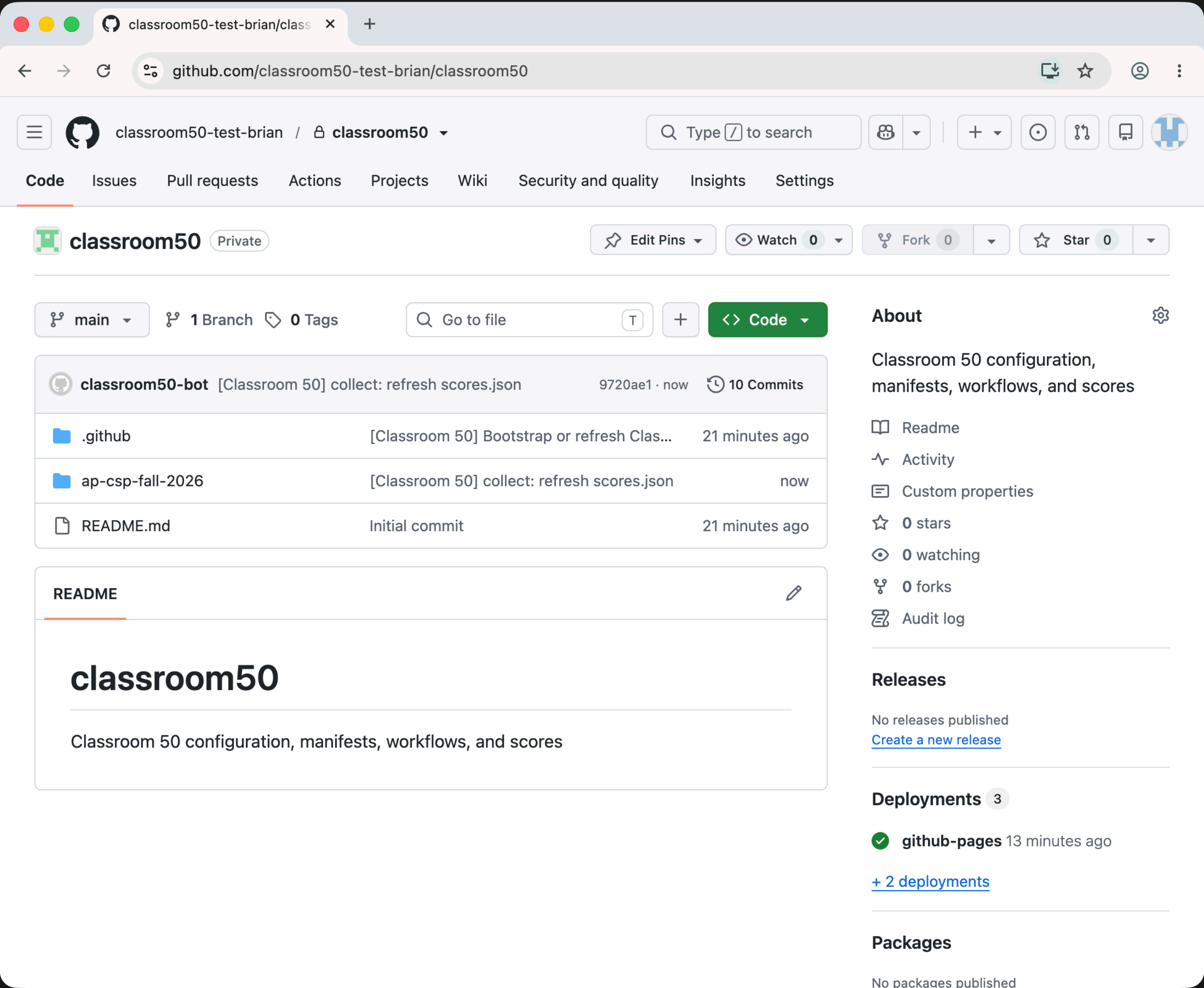
Interfaces



Web



Command-Line



GitHub

What You Need

- GitHub account
- GitHub organization with Team plan
 - Free via GitHub Education for verified teachers

GitHub Organization

Classroom 1

Assignment A

Assignment B

Assignment C

Classroom 2

Assignment D

Assignment E

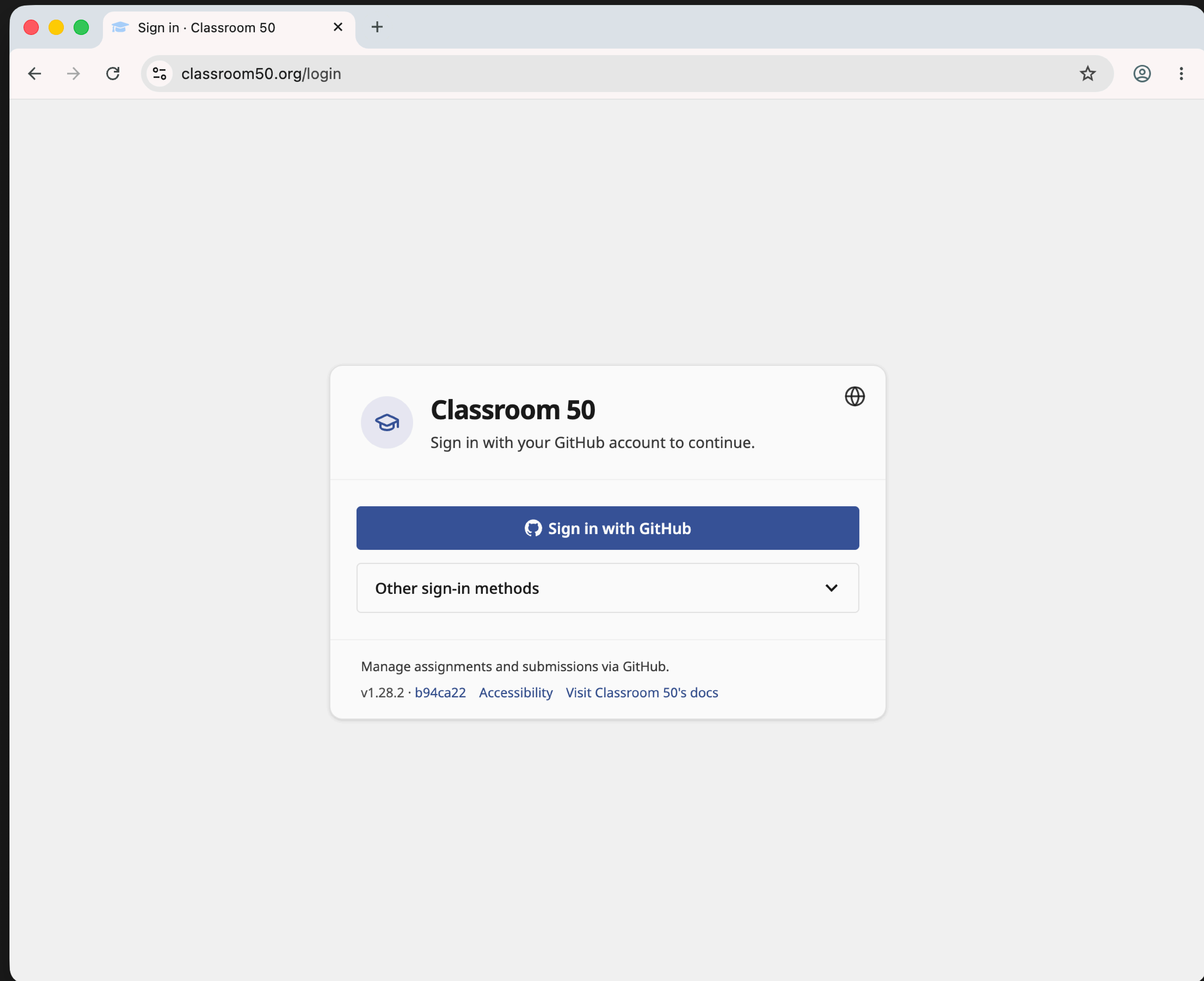
Classroom 3

Assignment F

Assignment G

Assignment H

Sign In



Organizations

Organizations · Classroom 50

classroom50.org

☆ ⬇️ 👤 ⋮

Classroom 50

Organizations

Classroom 50 organizations

Set up your first Classroom 50 organization

You own organizations that aren't set up yet. Configure Classroom 50 to start creating classrooms and assignments.

+ Set up an organization

Set up a new Classroom 50 organization

Pick one of your GitHub organizations to configure Classroom 50 in. Don't see the one you want? Create it on GitHub first.

Not seeing your organization?

Refresh list

ORGANIZATION READY TO SET UP (1)

classroom50-test-brian

Enterprise Plan

Set up

Create an organization on GitHub

Brian Yu

Setup

Setup · Classroom 50

classroom50.org/classroom50-test-brian/setup

☆ ⬇️ 👤 ⋮

Classroom 50 <

My classrooms

Published

Members

Activity

Settings

GITHUB ORGANIZATION

classroom50-test-brian

Brian Yu

Teacher

Setup Classroom 50

This will set up your GitHub organization to use Classroom 50.

1 2 3

Run setup

> Organization safety defaults

Locks down organization member privileges to safe defaults (base permissions, repo creation, visibility changes, and related settings).

> Actions permissions

Enables GitHub Actions for the organization.

> Actions spending cap

Creates a \$0 GitHub Actions spending cap (only if none exists).

> Actions pull request creation

Allows GitHub Actions to create and approve pull requests in the organization.

> Config repository

Creates the private classroom50 configuration repository in the organization.

> Skeleton files

Commits or updates the workflow and script files Classroom 50 needs in the classroom50 repository.

> Branch protection

Protects the classroom50 repository's main branch.

Create Classroom

Classroom 50

<

My classrooms

Published

Members

Activity

Settings

GITHUB ORGANIZATION

classroom50-test-brian

Brian Yu

Teacher

Classrooms > New classroom

Create classroom

Basic information

Classroom name *

AP Computer Science Principles

Classroom slug *

ap-csp-fall-2026

Classroom term

Fall 2026

☐

Use an unlisted link for this classroom

Publishes this classroom's assignment data at an unguessable URL instead of one anyone can reach by guessing the org name. This is obscurity, not real access control: anyone who gets the link can read it, and links can leak (browser history, referrers, search crawlers). Off by default.

Create classroom

Roster

Classroom 50

<

<

All classrooms

AP Computer Science Principles

Fall 2026

Assignments

Roster

Settings

GITHUB ORGANIZATION

classroom50-test-brian

Brian Yu

Teacher

Roster · Classroom 50

classroom50.org/classroom50-test-brian/ap-csp-fall-2026/roster

Classrooms > AP Computer Science Principles > Roster

Roster

5 members enrolled1 teacherView roster.csv

Search members

All members

All roles

5 members

+

↑

↗

BY

Brian Y.

@brianyu28

Teacher

>

AF

Anna F.

@student1

Student

>

DG

Dale G.

@student2

Student

>

AF

Sally R.

@student3

Student

>

AF

Shannon L.

@student4

Student

>

Create Assignment

Classroom 50

<

<

All classrooms

AP Computer Science Principles

Fall 2026

Assignments

Roster

Settings

GITHUB ORGANIZATION

classroom50-test-brian

Brian Yu

Teacher

New assignment · Classroom

+

classroom50.org/classroom50-test-brian/ap-csp-fall-2026/assignments/new

☆

👤

⋮

Classrooms > AP Computer Science Principles > New assignment

Create assignment

Details

Reset

Assignment name *

Fibonacci

Assignment slug * ?

fibonacci

Description (optional)

Describe the assignment objectives...

Assignment type

Individual

Group project

Max group size

2

Repository setup

Reset

Start with a template

No template

Create a fresh repository for each student.

Template repository

Generate each student's repository from a template you provide.

Repository features ↻

Whether each student repository has Wikis, Issues, Projects, and Pull requests enabled.

Configure Tests

New assignment · Classroom

classroom50.org/classroom50-test-brian/ap-csp-fall-2026/assignments/new

Classroom 50

← All classrooms

AP Computer Science Principles

Fall 2026

Assignments

Roster

Settings

GITHUB ORGANIZATION

classroom50-test-brian

Brian Yu

Teacher

Test name

Test 8

Test type

Input/Output

Run command

Python (pytest)

Run a command, feed it stdin, and compare its stdout against an expected value.

Setup command

optional – e.g., gcc -o hello hello.c

Run command

python3 fibonacci.py

Input (stdin)

8

Expected output

21

Comparison

Exact — output equals expected (surrounding whitespace ignored)

Add test

create assignment

Assignment

Submissions · Classroom 50

classroom50.org/classroom50-test-brian/ap-csp-fall-2026/assignments/fibonacci/submissions

Classroom 50

← All assignments

Fibonacci

Assignment

Submissions

Settings

Classrooms > AP Computer Science Principles > Assignments > fibonacci > Submissions

Fibonacci

No due date [View template](#)

Submission data synced 10 hours ago [Refresh submissions](#)

[Share](#)

[Actions](#)

Student	Submissions	Score	Last submitted	Actions
---------	-------------	-------	----------------	---------

How students accept

Share an invite link or CLI command so students can accept this assignment.

Sharing with [4 students](#) who can accept this assignment.

https://classroom50.org/classroom50-test-brian/ap-csp-fall-2026/as: [Copy](#)

> Prefer the command line?

GITHUB ORGANIZATION

classroom50-test-brian

Brian Yu

Teacher

Accept Assignment

Accept assignment · Classroo x

classroom50.org/classroom50-test-brian/ap-csp-fall-2026/assignments/fibonacci/accept

☆ ⓘ ⋮

 Classroom 50

🌐 Language

👤 Individual assignment

No due date

Fibonacci

Accept this assignment to get your own copy of the starter code repository.

Signed in as

 Anna F.
🔒 student1

Repository will be created as:

classroom50-test-brian/ap-csp-fall-2026-fibonacci-student1

Accept assignment

Student Repository

classroom50-test-brian/ap-cs

+

← → ↺

https://github.com/classroom50-test-brian/ap-csp-fall-2026-fibonacci-student1

🔒

👤 ⋮

☰

🐙

⋮

/

🔒 ap-csp-fall-2026-fibonacci-student1

▼

🔍 Type / to search

🔒 ▼

+

🕒

🔗

📄

🌐

Code

Pull requests

Actions

Security and quality

Insights

Settings

🏠

ap-csp-fall-2026-fibonacci-student1

Private

📌 Edit Pins ▼

👁 Watch 0 ▼

🔗 Fork 0 ▼

★ Star 0 ▼

🔗 main ▼

🔗 1 Branch

🏷 0 Tags

🔍 Go to file

T

+

<> Code ▼

🌐 student1

[Classroom 50] Initialize .classroom50.y...

⋮

●

ed69603 · 1 minute ago

🕒 2 Commits

📁 .github/workflows	[Classroom 50] Initialize .classroom50.ya...	1 minute ago
📄 .classroom50.yaml	[Classroom 50] Initialize .classroom50.ya...	1 minute ago
📄 fibonacci.py	Initial commit	1 minute ago

README

📖

Add a README

Help people interested in this repository understand your project.

Add a README

About

No description, website, or topics provided.

📶 Activity

📋 Custom properties

★ 0 stars

👁 0 watching

🔗 0 forks

📋 Audit log

Releases

No releases published

Create a new release

Packages

No packages published

Publish your first package

Contributors 1

🌐 student1

Anna F.

Autograding

Release Submission submit/2

https://github.com/classroom50-test-brian/ap-csp-fall-2026-fibonacci-student1/releases/tag/submit%2F2026-07-22T04-0...

...

ap-csp-fall-2026-fibonacci-student1

Type / to search

+ -

🔗

📄

🌐

Code

Pull requests 1

Actions

Security and quality

Insights

Settings

Releases

submit/2026-07-22T04-09-48Z-85d8c24

Submission submit/2026-07-22T04-09-48Z-85d8c24

Compare ▾

✎

🗑

Latest

github-actions

released this 3 minutes ago

submit/2026-...

85d8c24

✓

classroom50 autograde: 40/50

Test	Result	Score
Test 0	FAIL	0 / 10
Test 2	PASS	10 / 10
Test 3	PASS	10 / 10
Test 8	PASS	10 / 10
Test 10	PASS	10 / 10

► Failure details

Status: classroom50 autograde: 40/50 (4/5 tests passed)

▼ Assets

3

How It Works

- GitHub Organizations for **organizing submissions**
- GitHub Repositories for **storing classroom state**
- GitHub Pages for **publishing assignments**
- GitHub Actions for **autograding and collecting scores**
- GitHub Releases for **autograding results and feedback**

Walkthrough

CLI

Install GitHub CLI

- Classroom 50 ships as **GitHub CLI** extensions
- Install the GitHub CLI first at `cli.github.com`

Install Extensions



```
$ gh extension install foundation50/gh-teacher
```

```
$ gh extension install foundation50/gh-student
```


Log In



```
$ gh teacher login
```

```
$ gh student login
```

- Requests the scopes Classroom 50 needs
- Choose a browser (web) flow or a device code

Set Up Organization



```
$ gh teacher init <org>
```


Create a Classroom



```
$ gh teacher classroom add <org> <short-name>
```


Add a Student



```
$ gh teacher roster add <org> <classroom> <username>
```


Create an Assignment



```
$ gh teacher assignment add <org> <classroom> <slug> --name "<name>"
```


Accept an Assignment



```
$ gh student accept <org> <classroom> <assignment>
```


Submit Assignment



```
$ gh student submit
```


Collect Scores



```
$ gh workflow run collect-scores.yaml
```


View Scores



```
$ gh teacher download <org> <classroom> <assignment>
```


Recap

- **Teacher** sets up the org: `gh teacher init`
 - Create a classroom and add students: `classroom add, roster add`
 - Register an assignment: `assignment add`
- **Student** works on the assignment: `gh student accept, submit`
- **Teacher** grades: `collect-scores workflow`
- **Teacher** downloads scores with: `gh teacher download`

Q&A

Classroom 50

Teacher Training Session